Predicates and Quantifiers

Predicates

- The statement "x is greater than 3" has two parts. The variable x is the subject of the statement. "is greater than 3" is the **predicate**, which refers to a property that the subject of the statement can have.

- We denote the statement "x is greater than 3" by $P(x)$, where ***P* denotes the predicate "is greater than 3"** and ***x* is the variable**.

- The statement ***P*(*x*)** is said to be the value of the **propositional function** P at x.

- Once a value has been assigned to the variable x, the statement $P(x)$ becomes a proposition and has a truth value.

Example: Let $P(x)$: $x > 3$. What are the truth values of $P(4)$ and $P(2)$?

Solution:
$P(4)$: $4 > 3$ is True.
$P(2)$: $2 > 3$ is False.

We can denote the statement "$x = y + 3$" by $Q(x, y)$, where x and y are variables and Q is the predicate. When values are assigned to the variables x and y, the statement $Q(x, y)$ has a truth value.

Example: Let $Q(x, y)$: $x = y + 3$. What are the truth values of $Q(1, 2)$ and $Q(3, 0)$?

Solution:
$Q(1, 2)$: $1 = 2 + 3$ is False.
$Q(3, 0)$: $3 = 0 + 3$ is True.

We can let $R(x, y, z)$ denote the statement "$x + y = z$". When values are assigned to the variables x, y, and z, this statement has a truth value.

Example: Let $R(x, y, z)$: $x + y = z$. What are the truth values of $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution:
$R(1, 2, 3)$: $1 + 2 = 3$ is True.
$R(0, 0, 1)$: $0 + 0 = 1$ is False.

Quantifiers

- **Quantification** is a way of creating a proposition from a propositional function.

•The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

THE UNIVERSAL QUANTIFIER

- All possible values of a variable in a particular domain are called the **domain of discourse** or **universe of discourse** or the **domain**.
- The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain.

Definition 1: The *universal quantification* of $P(x)$ is the statement "$P(x)$ for all values of x in the domain." The notation $\forall xP(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the *universal quantifier*.

We read $\forall xP(x)$ as "for all x $P(x)$" or "for every x $P(x)$"

An element for which $P(x)$ is false is called a **counterexample** of $\forall xP(x)$.

Example: Let $P(x)$: $x + 1 > x$. What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Solution:
Because $P(x)$ is true for all real numbers x, the quantification $\forall x P(x)$ is true.

Example: Let $Q(x)$: $x < 2$. What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of real numbers?

Solution:
If $x = 3$, then $(x < 2) = (3 < 2)$ is not true (counterexample). Therefore, $\forall x Q(x)$ is false.

Example: Let $P(x)$: $x^2 > 0$. What is the truth value of the quantification $\forall x P(x)$, where the domain consists of integers?

Solution:
If $x = 0$, then $(x^2 > 0) = (0^2 > 0)$ is not true (counterexample). Therefore, $\forall x P(x)$ is false.

When all the elements in the domain can be listed, say $x_1, x_2, \ldots, x_n$, it follows that the universal quantification $\forall x P(x)$ is the same as the conjunction

$\forall x P(x) \equiv P(x_1) \land P(x_2) \land \ldots \land P(x_n)$

Example: What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement "$x^2 < 10$" and the domain consists of the positive integers not exceeding 4?

Solution:

Domain = {1, 2, 3, 4}

$\forall x P(x) \equiv P(1) \land P(2) \land P(3) \land P(4) \equiv \mathrm{T} \land \mathrm{T} \land \mathrm{T} \land \mathrm{F} \equiv \mathrm{F}$

Therefore, $\forall x P(x)$ is false.

Example: What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

Solution:

$x^2 \geq x$

or, $x^2 - x \geq 0$

or, $x(x - 1) \geq 0$.

Consequently, $x^2 \geq x$ if and only if $x \leq 0$ or $x \geq 1$.

If the domain consists of all real numbers,
$\forall x(x^2 \geq x)$ is false (since $x^2 \geq x$ is false for all real numbers x with $0 < x < 1$).

If the domain consists of integers,
$\forall x(x^2 \geq x)$ is true (since there are no integers x with $0 < x < 1$).

THE EXISTENTIAL QUANTIFIER

With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

Definition 2: The *existential quantification* of $P(x)$ is the proposition "There exists at least one element x in the domain such that $P(x)$." We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

Example: Let $P(x)$ denote the statement "$x > 3$." What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution:
Domain = All real numbers
Since $P(4)$: $4 > 3$ is true, $\exists x P(x)$ is true (example).

Example: Let $Q(x)$ denote the statement "$x = x + 1$." What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

Solution:
Domain: All real numbers
Since $Q(x)$ is false for every real number x, $\exists x Q(x)$ is false.

When all the elements in the domain can be listed, say $x_1, x_2, \ldots, x_n$, the existential quantification $\exists x P(x)$ is the same as the disjunction
$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n)$.

Example: What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement "$x^2 > 10$" and the domain consists of the positive integers not exceeding 4?

Solution:
Domain = {1, 2, 3, 4}
$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3) \vee P(4) \equiv \mathrm{F} \vee \mathrm{F} \vee \mathrm{F} \vee \mathrm{T} \equiv \mathrm{T}$
Therefore, $\exists x P(x)$ is true.

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.

For example, $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$.

Binding Variables

- When a quantifier is used on the variable x, we say that this occurrence of the variable is **bound**.
- An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be **free**.
- All the variables that occur in a propositional function must be bound or set equal to a particular value to turn it into a proposition.

- The part of a logical expression to which a quantifier is applied is called the **scope** of this quantifier.
- A variable is free if it is outside the scope of all quantifiers in the formula that specifies this variable.

Example:
In the statement $\exists x(x + y = 1)$, the variable x is bound by the existential quantification $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable.

Example:
In the statement $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$, all occurrences of the variable x are bound.
The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$.
The scope of the second quantifier, $\forall x$, is the expression $R(x)$.

Negating Quantified Expressions

Consider the proposition
"Every student in the class has taken a course in calculus."

This statement is a universal quantification $\forall x P(x)$
where $P(x)$ is the statement "x has taken a course in calculus."

The negation of this statement is
"There is at least one student in the class who has not taken a course in calculus."

This is simply the existential quantification of the negation of the original propositional function, $\exists x \neg P(x)$.

From this illustration we have the following equivalence:
$\neg \forall x P(x) \equiv \exists x \neg P(x)$.

Consider the proposition
"There is at least one student in this class who has taken a course in calculus."

This is the existential quantification $\exists x Q(x)$
where $Q(x)$ is the statement "x has taken a course in calculus."

The negation of this statement is the proposition
"Every student in this class has not taken a course in calculus,"

This is just the universal quantification of the negation of the original propositional function, $\forall x \neg Q(x)$.

This example illustrates the equivalence
$\neg \exists x Q(x) \equiv \forall x \neg Q(x)$.

Example: What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Solution:

$\neg\forall x(x^2 > x)$

$\equiv \exists x \neg (x^2 > x)$

$\equiv \exists x\ (x^2 \leq x)$.

$\neg\exists x(x^2 = 2)$

$\equiv \forall x \neg (x^2 = 2)$

$\equiv \forall x(x^2 \neq 2)$.

Example: Show that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution:

$\neg\forall x(P(x) \rightarrow Q(x))$

$\equiv \exists x(\neg(P(x) \rightarrow Q(x)))$

$\equiv \exists x(\neg(\neg P(x) \vee Q(x)))$

$\equiv \exists x(\neg(\neg P(x)) \wedge \neg Q(x))$

$\equiv \exists x(P(x) \wedge \neg Q(x))$

Nested Quantifiers

Nested quantifiers are quantifiers that occur within the scope of other quantifiers, such as in the statement $\forall x \exists y(x + y = 0)$.

Example:
Assume that the domain for the variables x and y consists of all real numbers.

The statement $\forall x \forall y(x + y = y + x)$ says that $x + y = y + x$ for all real numbers x and y (commutative law for addition of real numbers).

The statement $\forall x \exists y(x + y = 0)$ says that for every real numbers x there is a real number y such that $x + y = 0$, which states that every real number has an additive inverse.

The statement $\forall x \forall y \forall z(x + (y + z) = (x + y) + z)$ is the associative law for addition of real numbers.

The Order of Quantifiers

Example: Let $P(x, y)$ be the statement "$x + y = y + x$." What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ where the domain for all variables consists of all real numbers?

Solution:
The quantification $\forall x \forall y P(x, y)$ denotes the proposition
"For all real numbers x, for all real numbers y, $x + y = y + x$."

Because $P(x, y)$ is true for all real numbers x and y, the proposition $\forall x \forall y P(x, y)$ is true.

The statement $\forall y \forall x P(x, y)$ says
"For all real numbers y, for all real numbers x, $x + y = y + x$."

Because $P(x, y)$ is true for all real numbers y and x, the proposition $\forall y \forall x P(x, y)$ is true.

Example: Let $Q(x, y)$ denote "$x + y = 0$." What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of real numbers?

Solution:
The quantification $\exists y \forall x Q(x, y)$ denotes the proposition
"There is a real number y such that for every real number x, $Q(x, y)$."

Since there is no real number y such that $x + y = 0$ for all real numbers x, the statement $\exists y \forall x Q(x, y)$ is false.

The quantification $\forall x \exists y Q(x, y)$ denotes the proposition
"For every real number x there is a real number y such that $Q(x, y)$".

Given a real number x, there is a real number y (namely $y = -x$) such that $x + y = 0$. Hence, the statement $\forall x \exists y Q(x, y)$ is true.

Example: Let $Q(x, y)$ denote "$x + y = 0$." What are the truth values of the quantifications $\exists y \exists x Q(x, y)$ and $\exists x \exists y Q(x, y)$, where the domain for all variables consists of real numbers?

Solution:
The quantification $\exists y \exists x Q(x, y)$ denotes the proposition

"There is a real number y and a real number x such that $Q(x, y)$".

Given a real number y, there is a real number x (namely $x = -y$) such that $x + y = 0$. Hence, the statement $\exists y \exists x Q(x, y)$ is true.

The quantification $\exists x \exists y Q(x, y)$ denotes the proposition

"There is a real number x and a real number y such that $Q(x, y)$".

Given a real number x, there is a real number y (namely $y = -x$) such that $x + y = 0$. Hence, the statement $\exists x \exists y Q(x, y)$ is true.

Example: Let $Q(x, y, z)$ be the statement "$x + y = z$." What are the truth values of the statement $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consist of all real numbers?

Solution:
Suppose that x and y are assigned values. Then, there exists a real number z such that $x + y = z$. Thus, $\forall x \forall y \exists z Q(x, y, z)$ is true.

There is no value of z that satisfies the equation $x + y = z$ for all values of x and y. Thus $\exists z \forall x \forall y Q(x, y, z)$ is false.

Negating Nested Quantifiers

Example: Express the negation of the statement $\forall x \exists y(xy = 1)$ so that no negation precedes a quantifier.

Solution:

$\neg \forall x \exists y(xy = 1)$
$\equiv \exists x \neg \exists y(xy = 1)$
$\equiv \exists x \forall y \neg(xy = 1)$
$\equiv \exists x \forall y\ (xy \neq 1)$

Rules of Inference

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \rightarrow q$ $\therefore q$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\neg q$ $p \rightarrow q$ $\therefore \neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \rightarrow (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \rightarrow p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution